

SPPA: Scale-Adaptive Semantic Primitive Proxy Assembly for Real-Time UAV Digital Twins

Deep-AeroTwin Project

Abstract

Telemetry-driven UAV digital twins convert onboard perception into object-level messages that should be rendered as intended pilot-facing actors in a real-time virtual-reality interface. This creates a semantic telemetry-to-VR actor gap: the detector may provide a class, confidence, pose estimate, bounding region, or footprint, while the pilot needs an immediately visible object with meaningful occupied volume. Curated assets are incomplete, fixed-scale, and expensive to manage at runtime; high-fidelity image-to-3D models optimize a different contract. This paper specifies Semantic Primitive Proxy Assembly (SPPA), a bounded representation layer intended to compile semantic labels into low-polygon part graphs made of boxes, cylinders, ellipsoids, cones, and tori. The specification separates semantic compilation, primitive part graphs, measurement-conditioned pose/proportion fitting, uncertainty metadata, and conservative display fallback. SPPA is not a 3D reconstruction method and is not yet evaluated as a UAV/VR teleoperation system. It is a proposed operational proxy representation: the output is intended to preserve class family, approximate scale, pose, uncertainty, and occupied volume at predictable rendering cost. We provide the runtime contract, descriptor schema, algorithmic formulation, evidence-status table, and preregistered evaluation plan. The implemented artifact is limited to a debug prototype that generates seven fixed-template proxy classes through a batch/OBJ path in 60–80 ms per object. All claims about dense VR frame rate, end-to-end UAV telemetry integration, operator readability, conservative fallback behavior, and bandwidth reduction are marked as pending empirical validation.

1 Motivation

Pipeline B replaces continuous video transmission with lightweight semantic telemetry. The UAV does not transmit every pixel of the camera stream; it transmits object-level meaning: class label, confidence, position, timestamp, and approximate geometry. The ground station reconstructs the operational scene inside a Cesium-enabled Unreal Engine digital twin for a remote pilot wearing VR goggles.

This architecture creates a practical rendering problem. A detector may report many classes: people, cyclists, cars, trucks, animals, vegetation, poles, barriers, construction machinery, unknown obstacles, and rare objects. A digital twin cannot realistically maintain a complete, manually curated 3D asset library for every object that might appear during real-world UAV operation. Even if a large library is available, asset retrieval, licensing, semantic matching, level-of-detail management, and runtime loading remain operational problems.

Modern image-to-3D and text-to-3D models address a different objective. They are designed to create visually rich 3D assets from images or prompts. This is valuable for content creation, simulation authoring, games, and design. However, remote UAV piloting has a different priority: a detected object must appear in the pilot’s synthetic world almost immediately, with minimal polygon count and stable rendering cost. A visually impressive mesh generated several seconds later is less useful than a coarse obstacle volume generated in milliseconds.

2 Core Idea

The proposed system performs semantic primitive proxy generation. The key claim is simple:

For real-time UAV teleoperation, the digital twin does not need to reconstruct the true mesh of every detected object. It needs to reconstruct a low-latency, low-polygon, semantically recognizable volume that is sufficient for pilot situational awareness and obstacle avoidance.

Instead of producing a detailed mesh, the system assembles predefined geometric primitives into a proxy structure. The proxy is not only selected by class name; it is selected by a lightweight semantic and morphological interpretation of that class. If the detector reports `cow`, the system can infer that the object is an animal, a mammal, and a quadruped. That immediately constrains the proxy to include a body volume, a head volume, four legs, and an orientation hypothesis. If the detector reports `truck`, the system can infer a long vehicle body, a cab, wheels, and an elongated footprint. The silhouette then adjusts the proportions: a long truck and a short truck share the same semantic template, but the cargo body is scaled differently. For example:

- a cow may be represented by ellipsoids for body and head, cylinders for legs, and cones for horns;
- a cyclist may be represented by two torus wheels, a simple bicycle frame, and a human torso/head proxy;
- a tree may be represented by a cylinder trunk and clustered green volumes;
- a car or truck may be represented by boxes and wheel tori;
- a bush may be represented by overlapping green ellipsoids.

The exact geometry is intentionally crude. Its purpose is to preserve, pending empirical validation:

- approximate occupied volume;
- object class recognizability;
- spatial location;
- scale relative to the scene;
- extremely low rendering complexity.

An important property is that the proxy is scale-adaptive and proportion-adaptive. The system does not assume that all instances of a class have the same dimensions. A short truck, a long truck, a calf, an adult cow, a small bush, or a wide bush can share the same semantic part graph while receiving different part scales from the observed silhouette, bounding box, or estimated footprint. This is important for UAV piloting: when flying over an object, the relevant question is not whether the rendered actor is visually beautiful, but whether its occupied volume is approximately correct for navigation and operator interpretation. That claim remains a validation target until the planned volume-error and operator studies are completed.

3 Contributions

This manuscript is intentionally framed as a specification and validation plan. It separates what is specified, what is implemented, and what remains pending:

1. **Specified.** It defines *Semantic Primitive Proxy Assembly* (SPPA), a lightweight reconstruction layer that converts object-level UAV telemetry into class-conditioned geometric proxies.

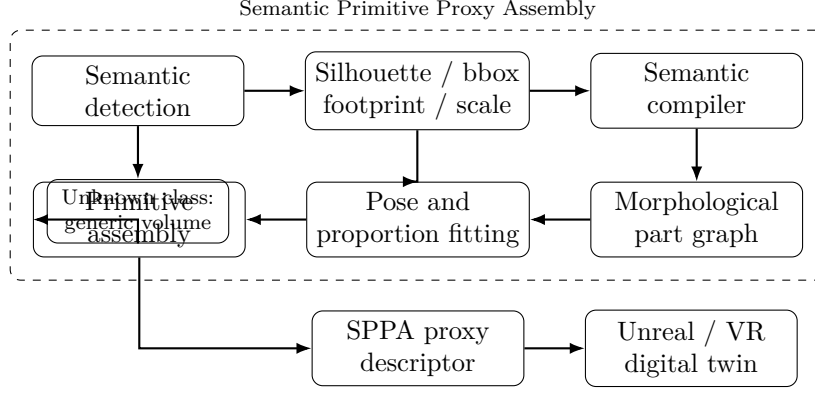


Figure 1: SPPA is specified to convert semantic telemetry into a low-polygon, proportion-adaptive proxy. The semantic compiler maps a detected label to a constrained morphological archetype and part graph, while the silhouette, bounding box, footprint, and temporal track constrain orientation, scale, and part proportions. The output is an operational volume for real-time VR rendering, not a high-fidelity mesh.

2. **Specified.** It proposes a bounded semantic-to-part compiler that maps detected labels to constrained morphological archetypes, semantic part graphs, and finite primitive geometry.
3. **Specified.** It introduces a scale-adaptive part-graph formulation in which a detected class activates a morphological prior, while silhouette or footprint evidence adapts the proportions of the parts.
4. **Positioned.** It positions primitive proxy assembly as a different operating point from neural image-to-3D generation: SPPA optimizes latency, polygon budget, proportional volume, and VR stability rather than visual fidelity.
5. **Implemented.** It reports a minimal debug prototype with seven fixed-template OBJ/MTL generators. This artifact demonstrates the visual style and debug-path generation timing only; it does not implement live telemetry, scale adaptation, descriptor-based Unreal spawning, or fallback behavior.
6. **Pending validation.** It defines experimental hypotheses and evaluation criteria for testing whether such proxies preserve enough operational information for remote UAV piloting under dense scene and degraded video-link conditions (PENDING-EXP, PENDING-BENCH, PENDING-USER).

4 Research Question and Hypotheses

The scientific question behind SPPA is not whether primitive geometry can look realistic. It is:

What is the minimum semantic-volumetric representation that preserves enough operational information for remote UAV piloting in a real-time VR digital twin?

This question leads to falsifiable hypotheses. None of H1–H5 is claimed as evaluated by the current prototype:

5 A Different Runtime Contract from 3D Generation

Recent 3D generative methods have advanced rapidly. DreamFusion introduced a text-to-3D optimization approach using a pretrained 2D diffusion model and score distillation sampling [23]. Magic3D improved quality and speed with a coarse-to-fine strategy, but still reports mesh creation

Table 1: Falsifiable SPPA hypotheses and current evidence status.

ID	Hypothesis to test	Required baselines	Primary metrics	Status
H1	SPPA is designed to reduce detection-to-visible-actor latency relative to runtime asset loading or neural generation.	box, capsule, low-poly asset load, image-to-3D	P50/P95 latency, triangles, descriptor bytes	PENDING-EXP
H2	SPPA is hypothesized to increase class-family recognition relative to generic boxes, capsules, or label billboards at comparable rendering cost.	3D box, capsule/ellipsoid, billboard label	recognition accuracy, response time, confusion matrix	PENDING-USER
H3	SPPA is designed to preserve footprint and occupied volume better than fixed-scale templates when object proportions vary.	fixed template, bbox-only scale, curated low-poly asset	BEV IoU, volume error, height error	PENDING-EXP
H4	SPPA is hypothesized to scale to dense VR scenes with more predictable rendering cost than detailed mesh assets.	boxes, capsules, low-poly assets, detailed meshes	FPS, CPU/GPU frame time, draw calls	PENDING-BENCH
H5	SPPA is designed to fall back to conservative display volumes under uncertain class, scale, silhouette, or yaw evidence.	no rendering, box, uncertainty volume	containment rate, false-confidence events, operator interpretation	PENDING-SAFETY

times on the order of tens of minutes [16]. Point-E reduced generation time substantially, producing point clouds in roughly 1–2 minutes on a single GPU, while explicitly trading quality for speed [19]. Shap-E further generates implicit 3D functions from text or image conditions in a matter of seconds [14].

More recent feed-forward methods are faster. InstantMesh reports diverse single-image 3D assets within approximately 10 seconds [31]. Stable Fast 3D reports textured mesh reconstruction from a single input image in approximately 0.5 seconds [26, 27]. TRELLIS and Hunyuan3D 2.0 focus on high-quality, versatile, textured 3D asset generation using large-scale latent or diffusion architectures [30, 34].

These methods are impressive, but they are not optimized for the same operating point. In UAV VR teleoperation, several constraints become dominant:

- **Per-object latency:** dynamic detections must appear in the digital twin immediately, not after seconds of neural inference.
- **Scene density:** the scene may contain hundreds of people, animals, vehicles, trees, or temporary obstacles.
- **VR frame rate:** the pilot interface must remain stable and comfortable; geometric complexity must be predictable.
- **Input availability:** a clean cropped image suitable for image-to-3D generation may not be available under poor visibility, motion blur, compression, occlusion, or low-confidence detection.
- **Operational sufficiency:** collision avoidance and situational awareness require approximate volume, not photorealistic surface detail.

The proposed primitive-proxy system therefore does not compete with neural asset generation in fidelity. It occupies a different region of the design space: millisecond-scale, low-polygon, class-conditioned volumetric representation.

Table 2: Operating point of SPPA compared with nearby representation choices.

Method family	Primary out-put	Optimized for	Limitation for UAV/VR
Bounding box	cuboid	speed	weak semantic readability
Curated asset	mesh actor	visual quality	incomplete library, fixed scale
Image-to-3D	detailed mesh	fidelity	latency and mesh complexity
3D detection	3D box/state	localization	not an intended pilot-facing actor
SPPA	part proxy	operational volume	coarse geometry

6 Relation to 3D Detection, Primitive Representations, and Human Factors

The proposed approach is related to several existing research directions. The claim is not that primitive part-based representation is new in isolation. The claim to be tested is narrower: SPPA is a proposed rendering/backend layer that turns semantic telemetry into bounded, low-polygon proxy actors for a UAV digital twin when exact assets or dense reconstruction are unavailable.

6.1 Monocular 3D Object Detection

Monocular 3D object detection predicts 3D properties such as object depth, dimensions, rotation, and oriented 3D bounding boxes from a single image. Omni3D introduced a large benchmark with 234k images, more than 3 million instances, and 98 categories, together with Cube R-CNN as a unified detector across diverse camera and scene types [3]. Recent work also studies open-vocabulary monocular 3D detection, aiming to lift 2D detections into metric 3D space beyond closed class vocabularies [32].

These methods are highly relevant because they can provide exactly the parameters that semantic primitive proxies need: approximate depth, orientation, and dimensions. However, their output is normally a 3D bounding cuboid or object state, not a semantically readable, part-based VR actor. Our proposal can use monocular 3D detection as an upstream estimator, then convert the detected class and dimensions into a low-polygon operational proxy.

6.2 Animal Pose and Animal 3D Reconstruction

For animals, the state of the art includes keypoint-based pose estimation and model-based reconstruction. AP-10K provides a benchmark for mammal pose estimation across 23 animal families and 54 species [33]. 3D-Fauna reconstructs articulated 3D meshes of quadruped animals from a single image within seconds using a pan-category animal model [15]. More recent work such as SAM 3D Animal uses prompts such as masks and keypoints to reconstruct multiple animals in the wild using SMAL-based priors [12].

These systems demonstrate that richer animal reconstruction is possible. The operational question is whether such reconstruction is necessary for UAV teleoperation. For a pilot, the immediate need is often not fur texture, exact limb articulation, or animatable mesh quality. The immediate need is: a large quadruped occupies this volume, its body is oriented in this direction, and this space should be avoided. A primitive proxy can represent that information with substantially lower geometric and runtime cost.

6.3 Primitive and Superquadric Decomposition

Representing objects as combinations of simple shapes is a long-standing idea in computer vision and robotics. Superquadrics are especially relevant because they compactly represent boxes, cylinders, spheres, ellipsoids, and related forms with few parameters. Earlier shape-representation work used generalized cylinders and volumetric components to describe object structure [17], and recognition-by-components proposed that objects can be recognized from arrangements of simple geon-like parts [2]. Superquadric recovery and parsing further developed compact primitive models for object abstraction [25, 22]. Learned volumetric primitive abstraction also shows that primitive assemblies can capture consistent part structure across shape collections [28], while recent systems such as SuperDec explicitly use superquadric primitives as a compact representation for 3D scene decomposition [9].

This literature is a direct prior, not a minor analogy. SPPA must therefore be judged as an operational telemetry-to-rendering contract rather than as a new theory of primitive shape representation. The remaining open question is whether a constrained primitive representation is useful in the specific Pipeline B setting, compared with simpler boxes/capsules and with existing asset or mapping representations.

6.4 Procedural Actors, LOD, and Scene Graphs

SPPA is also related to procedural modeling, shape grammars, level-of-detail systems, impostors, semantic scene graphs, and bounding-box visualization. These are closer baselines than slow text-to-3D systems. Procedural modeling and CGA shape grammars generate structured geometry from compact rules [18]. ShapeAssembly similarly uses executable programs to assemble cuboid part proxies and represent families of related shapes [13]. Against this literature, SPPA’s proposed difference is not procedural generation itself; it is the use of a bounded procedural descriptor as an online telemetry rendering fallback.

Real-time graphics has long used hierarchical level of detail, impostors, and extreme simplification to control rendering cost [4, 5]. Modern engines and geospatial standards continue this line through virtualized geometry, instancing, and tiled streaming representations [8, 21]. SPPA must therefore be benchmarked against low-poly assets, LOD, impostors, and instanced primitive rendering rather than only against neural 3D asset generators.

Semantic scene graphs are another direct neighbor. 3D scene graphs and dynamic scene graphs connect objects, semantics, geometry, dynamics, and actionability [1, 24]. SceneGraphFusion shows that incremental 3D semantic graph prediction can run online from RGB-D sequences [29]. SPPA can be interpreted as a rendering/backend layer for such semantic state, not as a replacement for scene-graph perception. This framing reduces the novelty claim but makes the system boundary clearer.

6.5 Human Factors, Display Uncertainty, and Mapping Baselines

Because the intended output is for a remote pilot, SPPA cannot rely on visual plausibility alone. Situation awareness and workload should be evaluated with established human-factors instruments such as SAGAT and NASA-TLX [6, 7, 10]. Any claim about operator readability, obstacle-avoidance usefulness, or workload reduction is therefore marked PENDING-USER until controlled operator studies are performed.

SPPA also overlaps with robotic mapping representations. Occupancy grids, OctoMap, TSDFs, and ESDFs provide spatial volumes and obstacle-distance information for planning and navigation [11, 20]. These representations are not intended to be semantic proxy actors by themselves, but they are essential baselines for any claim about operational volume. The publishable evaluation must therefore distinguish pilot-facing semantic readability from collision or planning utility.

The closest practical comparison is therefore not “SPPA versus high-fidelity 3D generation,” but SPPA versus no rendering, 3D boxes, capsules/ellipsoids, billboard labels, curated low-poly assets,

procedural actors, occupancy-style volumes, and instanced rendering in the target Unreal/VR environment.

7 Operation Inside Pipeline B

SPPA is intended to run at the ground station or inside the rendering component of the digital twin. The onboard UAV system detects objects and transmits semantic telemetry rather than full video frames. The ground station receives this telemetry, updates tracked entity states, and invokes SPPA only when an entity must be created, updated, or reparameterized.

The runtime loop is:

1. The UAV detects an object and estimates its class, confidence, image region, and geospatial position.
2. The ground station associates the detection with an entity track.
3. SPPA selects or updates a semantic part graph for that entity.
4. The silhouette, footprint, and temporal track adjust orientation, scale, and part proportions.
5. Unreal Engine instantiates or updates primitive components using the resulting proxy descriptor.
6. The pilot sees a stable, low-polygon operational representation of the object in VR.

This loop separates perception from visual fidelity. If a high-quality asset is available and affordable, it may be used. If not, SPPA is designed to ensure that the entity still receives a proportional, semantically meaningful obstacle volume.

8 Proposed Runtime Contract

The system is designed as a fallback or primary lightweight rendering layer inside Pipeline B. Its inputs are semantic detections and geometric estimates from the perception pipeline. Its outputs are low-polygon proxy actors spawned in the digital twin.

Table 3: Semantic primitive proxy runtime contract.

Stage	Input	Output
Detection	image frame	class, confidence, silhouette/bounding box
Georeferencing	UAV pose + camera model	world position and scale estimate
Proxy selection	object class	primitive template or generic fallback
Primitive scaling	silhouette/footprint/volume	scaled proxy geometry
Actor spawning	proxy mesh + transform	Unreal digital twin actor
Runtime update	tracked entity state	pose, scale, confidence, uncertainty

The expected telemetry fields are:

- stable entity identifier;
- detected class label;
- confidence score;
- geospatial position;
- estimated footprint, height, or bounding volume;

- silhouette or 2D bounding box when available;
- timestamp and sensor source;
- uncertainty estimate.

Table 4: Evidence status for the SPPA manuscript.

Claim or component	Current evidence	Status
Seven fixed proxy templates	Implemented in the debug OBJ/MTL generator and timed through the batch path.	Implemented
SPPA descriptor contract	Specified as a compact part/pose/uncertainty descriptor; not yet deployed as an Unreal runtime API.	Specified only (PENDING-IMPL)
Scale- and footprint-conditioned fitting	Defined algorithmically, but not implemented in the current prototype.	Specified only (PENDING-SPEC)
Unknown-class conservative display fallback	Specified as required behavior; the current prototype exits on unsupported labels.	Pending empirical validation (PENDING-SAFETY)
End-to-end UAV-to-VR loop	No live YOLO tracks, camera calibration, georeferenced footprints, or Unreal actor spawning are evaluated here.	Pending implementation (PENDING-IMPL)
Dense VR performance	No headset, Unreal, CPU/GPU, draw-call, or FPS benchmark is reported.	Pending empirical validation (PENDING-BENCH)
Operator readability and situational awareness	No pilot or user study is reported.	Pending empirical validation (PENDING-USER)
Bandwidth reduction	No comparison with encoded video, telemetry packet rates, retransmission, or full-mesh transfer is reported.	Pending empirical validation (PENDING-EXP)

9 SPPA Algorithm

SPPA is designed as a deterministic, low-latency algorithm. It should not call a large generative model at runtime, and it should not search a large asset library. Its task is to transform the minimum available semantic and geometric evidence into an operational 3D proxy. The algorithm can be summarized as:

1. **Normalize the semantic label.** The detector output is mapped to a semantic path such as `cow` $\rightarrow$ `animal` $\rightarrow$ `mammal` $\rightarrow$ `quadruped` or `truck` $\rightarrow$ `vehicle` $\rightarrow$ `cargo vehicle`.
2. **Select a morphological archetype.** The semantic path selects a small family of part graphs, such as `quadruped`, `biped`, `long vehicle`, `vegetation blob`, `tree-like`, `pole-like`, or `generic obstacle`.
3. **Extract geometric measurements.** The system estimates major and minor footprint axes, aspect ratio, height, ground contact, and yaw candidates from the bounding box, mask, footprint, previous track, and UAV camera geometry.
4. **Fit pose and proportions.** The algorithm chooses a yaw angle and assigns length, width, and height to semantic parts while respecting class-level priors and observed silhouette proportions.

5. **Assemble primitives.** Each semantic part is instantiated as a low-resolution primitive: box, cylinder, ellipsoid, cone, torus, capsule, or superquadric-like approximation.
6. **Estimate uncertainty and fallback.** If class, orientation, or scale is ambiguous, the output includes uncertainty metadata or falls back to a conservative generic volume.

9.1 Semantic Normalization

SPPA should support many class labels without requiring a manually authored asset for each one. The label is therefore mapped to a compact ontology of morphological archetypes. The ontology does not need to be large; it only needs to capture the body plan relevant for operational volume:

Table 5: Examples of semantic normalization and archetype selection.

Detected label	Semantic path	Archetype
cow, horse, dog	animal / mammal / quadruped	quadruped
person, pedestrian	human / biped	biped
biker, cyclist	human + two-wheel vehicle	rider vehicle
car, van	vehicle / wheeled	compact vehicle
truck, bus	vehicle / long wheeled	long vehicle
tree	vegetation / trunk + canopy	tree-like
bush, shrub	vegetation / low blob	vegetation blob
pole, mast, tower	vertical structure	pole-like
unknown	obstacle	generic volume

9.2 Bounded Semantic-to-Part Compiler

The important architectural step in SPPA is the conversion from a detected semantic label to a constrained geometric part graph. This can be viewed as a semantic compiler. The input vocabulary may be open-ended, but the output is restricted to a finite set of morphological archetypes, semantic roles, and primitive geometries.

Let c be the detected label. A language model, knowledge graph, or manually curated ontology maps c to a semantic descriptor:

$$z_c = \mathcal{L}(c) = (h_c, a_c, q_c, d_c),$$

where h_c is a category hierarchy, a_c is a set of expected attributes, q_c is a set of qualitative morphology cues, and d_c contains coarse dimension priors. For example, a descriptor for **cow** may include **animal**, **mammal**, **quadruped**, **has head**, **has torso**, and **four legs**; a descriptor for **truck** may include **vehicle**, **wheeled**, **cab**, **cargo body**, and **elongated footprint**.

The semantic descriptor selects the closest morphological archetype:

$$A^*(c) = \arg \max_{A \in \mathcal{A}} \text{sim}(\phi(A), z_c) \quad \text{s.t.} \quad \text{risk}(A, z_c) \leq \rho_{\max},$$

where $\mathcal{A}$ is a compact library of archetypes, $\phi(A)$ is the semantic signature of an archetype, and the risk constraint is a `PENDING-SPEC` validation rule that should prevent over-specific mappings when semantic evidence is weak. The functions `sim`, `risk`, `descriptor calibration`, and threshold $\rho_{\max}$ are not fixed by the current prototype; they must be specified before claiming open-vocabulary coverage. If no archetype is sufficiently reliable, the compiler selects a conservative display obstacle.

The selected archetype is expanded into a semantic part graph:

$$G_c = \Gamma(A^*, z_c) = (V_c, E_c),$$

with nodes

$$v_i = (r_i, \omega_i, \pi_i, \alpha_i, \kappa_i).$$

Here r_i is the semantic role of the part, ω_i is the primitive type selected from a finite primitive alphabet $\Omega = \{\text{box, cylinder, ellipsoid, cone, torus, capsule}\}$, π_i is the anchor relative to the parent part, α_i stores allowed aspect-ratio ranges, and κ_i stores importance and uncertainty metadata. Edges E_c encode attachment and relative placement constraints, such as head-attached-to-torso, wheels-attached-to-chassis, or canopy-above-trunk.

Finally, the graph is instantiated as geometry:

$$\mathcal{P}(G_c, \theta) = \bigcup_{v_i \in V_c} T(\pi_i, \theta_i) \omega_i(\alpha_i),$$

where θ_i contains the fitted pose and scale parameters of each part. This equation is the architectural core of SPPA: arbitrary semantic labels are not converted directly into arbitrary meshes; they are compiled into a bounded, low-polygon graph of semantic parts drawn from a finite primitive vocabulary.

This gives SPPA bounded semantic coverage. Any label that can be mapped to a morphological descriptor can be represented by a proxy, even if no exact asset exists. The quality of the proxy depends on the quality of the semantic descriptor and visual evidence, but the system always degrades toward a conservative display volume rather than silently dropping the object.

9.3 Part Graphs

Each archetype is represented by a semantic part graph. Nodes are parts and edges encode relative placement constraints. A node contains:

- semantic role, such as torso, head, leg, cab, cargo body, wheel, trunk, canopy, or shaft;
- primitive type, such as box, ellipsoid, cylinder, cone, or torus;
- relative anchor and transform;
- allowed aspect-ratio and scale range;
- recognition importance;
- polygon budget.

For example, a quadruped graph contains a body ellipsoid, a head ellipsoid, a short neck cylinder, four leg cylinders, and an optional tail. A long-vehicle graph contains a cargo box, cab box, wheels, and optional windshield marker. The same graph can represent many real dimensions because part scales are not fixed.

9.4 Measurement Extraction

The system can operate with different levels of geometric evidence:

- **Footprint available:** fit a minimum-area oriented rectangle and extract length, width, aspect ratio, and yaw candidates.
- **Silhouette mask available:** use principal axes, asymmetry, protrusions, and lower contact regions to refine orientation and part allocation.
- **Bounding box only:** combine image-space extent with class priors, UAV pose, and ground-plane assumptions to estimate a conservative volume.

- **Temporal track available:** use velocity and previous yaw to reduce front/back ambiguity and jitter.

This makes the method applicable under degraded sensing. A mask can improve the proxy, but the algorithm still produces a useful volume with only a class label and a bounding region.

9.5 Conservative Fallback

The SPPA specification requires a usable display volume, even when the class is unknown or the silhouette is too weak for reliable part assignment. The fallback is selected from simple geometric evidence:

- elongated and low footprint: oriented capsule or elongated box;
- small footprint and large height: pole-like cylinder;
- low circular footprint: vegetation blob or generic ellipsoid;
- upright human-scale region: biped proxy or vertical capsule;
- ambiguous region: conservative oriented bounding volume.

The fallback is not treated as a successful reconstruction. It is a conservative display mechanism intended to preserve operational continuity when semantic detail is missing. The operator sees a volume with uncertainty, and the digital twin does not silently drop an obstacle. Whether this reduces risk or creates false confidence is a PENDING-SAFETY validation question.

9.6 Pose and Proportion Fitting

SPPA estimates orientation and dimensions by combining weak cues, but the yaw estimate must distinguish axial and directional evidence. Footprint principal axes and silhouette PCA normally provide an unoriented axis modulo π , while velocity, asymmetric part evidence, or a previous signed track yaw can provide a direction modulo 2π . The implementation therefore first estimates an axial orientation

$$\bar{\psi}_t^{\text{axis}} = \frac{1}{2} \text{atan2} \left(\sum_k w_k \sin 2\psi_k^{\text{axis}}, \sum_k w_k \cos 2\psi_k^{\text{axis}} \right),$$

and then assigns the front/back sign only when directional evidence is available:

$$(\psi_t, \text{yaw_ambiguous}) = \text{orient} \left(\bar{\psi}_t^{\text{axis}}, \{\psi_j^{\text{dir}}, w_j\}_j \right).$$

For vehicles, the cabin or velocity may identify the front. For quadrupeds, a head-like protrusion or motion direction may identify the head. If the evidence is weak, `yaw_ambiguous = true` and the proxy remains a symmetric obstacle volume. The exact confidence thresholds for orient are

PENDING-SPEC.

Part dimensions are fitted analytically when possible. For a long vehicle:

$$L_{\text{cargo}} = \alpha L, \quad L_{\text{cab}} = (1 - \alpha)L, \quad \alpha \in [0.55, 0.80],$$

where L is the measured footprint length. Thus a short truck and a long truck are represented by the same graph but different cargo-body scale. For a quadruped:

$$L_{\text{body}} = \beta L, \quad L_{\text{head}} = (1 - \beta)L, \quad H_{\text{leg}} = \lambda H,$$

with β and λ constrained by the quadruped prior and by the observed silhouette. In an implementation, the fitting can be performed by a small discrete hypothesis search over 8–32 candidates rather than a heavy optimization loop.

9.7 SPPA Objective

The full SPPA target is specified as a finite hypothesis selection problem, not as an expensive continuous optimizer. Given a detection D_t , an archetype A , measurements such as footprint F_t , optional mask M_t , and previous track state x_{t-1} , SPPA constructs a small candidate set $\mathcal{H}(A, F_t, M_t, x_{t-1})$. Each candidate separates global pose $\xi_t = (p_t, \psi_t)$ from part parameters q_t , including part scales, local anchors, and fallback mode. The selected proxy is:

$$(\xi_t^*, q_t^*) = \arg \min_{(\xi, q) \in \mathcal{H}(A, F_t, M_t, x_{t-1})} w_F (1 - \text{IoU}_{BEV}(P_{\xi, q}, F_t)) + w_S (1 - \text{IoU}_{mask}(\Pi(P_{\xi, q}), M_t)) \\ + w_T d_{SE(2)}(\xi, \xi_{t-1}) + w_A \ell_A(q)$$

$$\text{s.t.} \quad \text{tri}(P_{\xi, q}) \leq N_{\max}, \quad \hat{\tau}(P_{\xi, q}; E, H) \leq \tau_{\max}.$$

Here $P_{\xi, q}$ is the candidate primitive proxy, IoU_{BEV} measures footprint overlap in bird’s-eye view, IoU_{mask} measures optional image-mask agreement, $d_{SE(2)}$ penalizes pose jitter, and ℓ_A penalizes proportions outside the selected archetype’s allowed ranges. If M_t is unavailable, $w_S = 0$; if no reliable footprint exists, SPPA uses class priors and conservative fallback volumes. The runtime term $\hat{\tau}$ is explicitly hardware- and engine-dependent: E denotes the rendering engine/runtime path and H denotes target hardware. It cannot be proven from triangle count alone and must be measured in Unreal/VR (PENDING-BENCH). The weights w_F, w_S, w_T, w_A , normalization of objective terms, and candidate-generation policy are PENDING-SPEC.

9.8 Role of a Language Model

A language model is not required to generate geometry at frame rate. In SPPA, its role is better understood as a semantic compiler or cache builder. When a new label appears, the language model can propose a descriptor z_c , a parent archetype, and a minimal part graph. That proposal is then reviewed against the finite primitive vocabulary, polygon budget, and conservative display rules of SPPA. Once reviewed and accepted, the mapping can be cached and reused deterministically. The review procedure, prompt/model version, cache contents, and rejection criteria are PENDING-SPEC.

This distinction matters for real-time UAV operation. The runtime loop should not depend on a slow or nondeterministic text-to-geometry call. The language model helps extend the open vocabulary; the real-time system still executes a bounded geometric program. For example, if the detector reports `camel`, the compiler may map it to a quadruped with a larger torso, long neck, and optional hump. If it reports `excavator`, it may map it to a vehicle-like body with tracks, cabin, and arm proxy. If the proposal is uncertain, SPPA falls back to the nearest parent archetype or to a conservative generic volume.

The coverage claim is therefore conditional and operational, not photorealistic. SPPA can generate a useful geometric proxy for many arbitrary semantic labels if those labels can be mapped to a morphological descriptor or a conservative fallback. It does not guarantee an exact mesh for every object in the world, and it does not yet establish reliable open-vocabulary behavior. This distinction is central: the method specifies bounded semantic proxy coverage, not universal 3D reconstruction.

9.9 Runtime Output

At runtime, the preferred output is not an OBJ mesh. OBJ export is useful for debugging and paper figures, but a deployed Unreal system should receive a compact proxy descriptor that can be rendered using instancing or procedural mesh components:

```
{
  "label": "cow",
  "archetype": "quadruped",
```

```

"pose": {"position": [x, y, z], "yaw": psi},
"parts": [
    {"role": "body", "type": "ellipsoid", "scale": [2.1, 0.7, 0.8]},
    {"role": "head", "type": "ellipsoid", "scale": [0.45, 0.35, 0.35]},
    {"role": "leg", "type": "cylinder", "count": 4}
],
"uncertainty": {"scale": 0.22, "yaw_deg": 35, "semantic": 0.81}
}

```

This descriptor is small enough to transmit, cache, smooth over time, and render at large scale.

9.10 Pseudocode

```

function SPPA(detection, track_state, ontology, archetypes):
    label = normalize_label(detection.class_label)
    semantic_path = ontology.resolve(label)
    measurements = extract_measurements(detection)

    archetype = select_archetype(semantic_path, measurements)
    part_graph = archetypes[archetype]

    pose = fit_pose(
        measurements,
        previous = track_state[detection.track_id],
        semantic_path = semantic_path
    )

    dimensions = fit_dimensions(
        part_graph,
        measurements,
        confidence = detection.confidence
    )

    params0 = analytic_initialization(part_graph, dimensions)
    params = small_hypothesis_search(
        part_graph,
        params0,
        measurements,
        pose,
        max_candidates = 32
    )

    parts = instantiate_primitives(part_graph, pose, dimensions, params)
    parts = enforce_polygon_budget(parts, detection.distance_to_pilot)

    uncertainty = estimate_uncertainty(
        detection, measurements, pose, dimensions, semantic_path
    )

    if uncertainty.too_high:
        parts = conservative_fallback_volume(measurements, semantic_path)

    proxy = make_proxy_descriptor(label, semantic_path, pose, parts, uncertainty)

```

```
track_state.update(detection.track_id, proxy)
return proxy
```

10 Part-Fitting Examples

This section illustrates how the SPPA specification is applied to two common object families. The examples are not special cases; they show how a class prior and observed geometry jointly determine a part graph and its proportions.

The class prior is modeled as a semantic part graph. A class label activates both a category hierarchy and a minimal part structure. For instance:

`cow → animal → mammal → quadruped`

From this hierarchy the system can infer that a proxy should contain at least a body, a head, and four leg supports. Similarly:

`truck → vehicle → {cab, cargo body, wheels}`

This makes the reconstruction compositional. The system is not generating an arbitrary mesh; it is fitting a small semantic graph of geometric parts to the detected silhouette and estimated world scale.

Let a detection be represented as:

$$D_t = (c, p, B, s, \gamma, t)$$

where c is the detected class, p is the estimated world position, B is the image-space bounding box or silhouette, s is an approximate world scale, γ is confidence, and t is timestamp. The proxy generator selects a primitive template:

$$T_c = \{(g_i, r_i, m_i)\}_{i=1}^{n_c}$$

where g_i is a primitive geometry type, r_i is its relative transform within the object, and m_i is its semantic material. More explicitly, each primitive belongs to a semantic part:

$$T_c = \{(a_i, g_i, r_i, \theta_i, m_i)\}_{i=1}^{n_c}$$

where a_i is a part label such as torso, head, leg, cab, cargo body, or wheel, and θ_i contains part-specific scale limits and deformation rules. The final proxy separates global pose, local part transform, and primitive dimensions:

$$P_t = \bigcup_{i=1}^{n_c} T_{\text{world}}(p_t, \psi_t) T_i(a_i, \delta_i) \omega_i(q_i),$$

where T_{world} places the object in the digital twin, T_i places part i relative to its parent, ω_i is the selected primitive, and q_i contains the fitted primitive dimensions.

The silhouette is used to estimate a coarse yaw angle and to allocate scale to the most likely parts. For a quadruped, the largest elongated region is treated as the torso candidate; smaller protrusions near one end suggest head position; thin lower protrusions suggest legs. For a truck, the elongated body region sets the cargo body length, while a shorter high-volume region can become the cab. In ambiguous cases, the proxy should preserve uncertainty rather than claiming a precise pose.

This proportional fitting is one of the central differences between a fixed asset lookup and a semantic primitive proxy. A conventional asset library may contain a single truck mesh or several manually authored variants. A neural generator may produce a plausible truck mesh, but it is not necessarily conditioned on the real detected footprint in a way that enforces a low-polygon, VR-stable obstacle volume. The primitive proxy instead treats the detected silhouette as a constraint

on part proportions. The same template can therefore represent both a short utility truck and a long cargo truck by stretching the cargo body while preserving cab and wheel semantics.

This formulation makes three design choices explicit:

1. the proxy is conditioned by semantic class and category hierarchy;
2. the proxy is decomposed into a small graph of semantic parts;
3. the proxy is oriented and scaled by the detected silhouette or estimated footprint;
4. the geometry is intentionally constrained to a small number of primitives.

If the class is unknown, the system can fall back to a generic uncertainty volume, such as a translucent bounding box, ellipsoid, or vertical cylinder. This is operationally preferable to failing to render an object.

11 Implementation Status

To avoid overstating the current prototype, Table 6 separates implemented functionality from the full SPPA architecture.

Table 6: Current prototype versus full SPPA architecture.

Component	Current prototype	Full SPPA target
Semantic input	Single text label entered via batch script	Detection message with label, confidence, track ID, pose, bbox, mask, footprint, and uncertainty
Class handling	Seven manually authored templates; unsupported labels abort in the current script	Bounded semantic compiler with archetypes, reviewed descriptors, cache, and conservative display fallback
Geometry scaling	Fixed class-level proportions	Footprint-, silhouette-, and track-conditioned part scaling
Orientation	Template default orientation	Fused yaw from footprint, silhouette, velocity, and previous track
Runtime output	OBJ/MTL files for visual debugging	Compact primitive descriptor for instanced Unreal rendering
Evaluation	Per-object generation timing through batch/OBJ path	Latency, polygon budget, dense-scene FPS, footprint/volume error, recognition, and pilot-task metrics

12 Prototype

We implemented a minimal procedural prototype named **XYT-xabi-yolo-telemetry**. The prototype receives a word from a Windows batch file and exports an OBJ/MTL mesh using only procedural geometry. The current supported classes include:

- `cow`;
- `biker`;

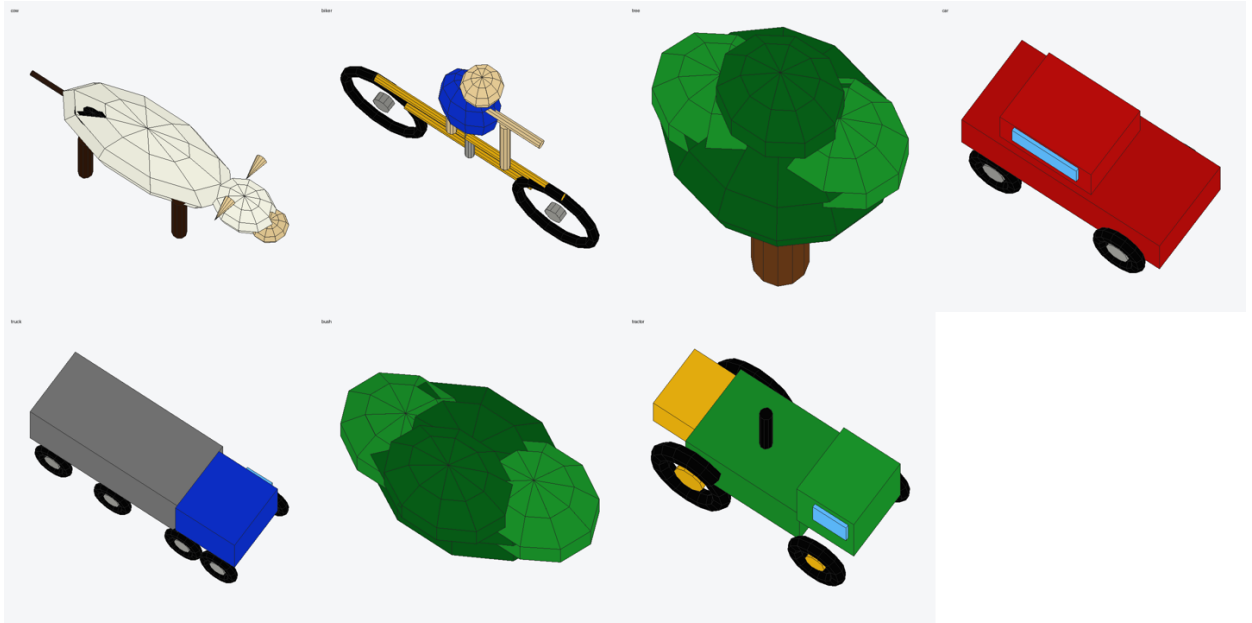


Figure 2: Example primitive proxies generated by the current prototype. The figures are deliberately low fidelity: their role is to communicate class, approximate orientation, and occupied volume with minimal geometry. In a runtime Unreal implementation, these proxies should be instantiated from primitive descriptors rather than loaded as high-detail meshes.

- `tree;`
- `car;`
- `truck;`
- `bush;`
- `tractor.`

This prototype is intentionally simple. It demonstrates the speed and geometry style of the approach, not a full UAV perception integration. In the current implementation, each class is mapped to a fixed lightweight primitive template. The prototype does not consume detections, tracks, masks, camera calibration, georeferenced footprints, or uncertainty. It also does not implement the unknown-class conservative display fallback specified above; unsupported labels currently terminate with an error. In the full Pipeline B target (`PENDING-IMPL`), templates would be scaled using the detected silhouette, estimated depth, or geospatial footprint.

13 Prototype Timing and Geometry Complexity

The prototype was measured by invoking the actual batch interface used by the operator. The observed wall-clock generation times and exported geometry sizes were:

These values include Python process startup and file export overhead. They are reported only as debug-path observations. They do not establish the latency of a descriptor-based Unreal implementation, dense-scene rendering, headset frame rate, or detection-to-visible-actor latency. Those measurements are `PENDING-BENCH`. The relevant prototype result is modest: seven fixed low-polygon templates can be exported through the current batch/OBJ path in under 0.1 s per object.

14 Operational Claim To Be Tested

The scientific contribution is not a new high-fidelity 3D generator. The testable claim is instead:

Table 7: Prototype OBJ generation time and geometry complexity. These values measure the observed debug path only, not a deployed Unreal instancing runtime and not an end-to-end UAV/VR path.

Class	Time (s)	Vertices	Faces	OBJ bytes
biker	0.061	488	396	21,263
bush	0.081	256	210	10,888
car	0.062	448	408	20,268
cow	0.065	612	502	26,533
tractor	0.074	500	450	22,629
tree	0.062	288	236	12,308
truck	0.076	856	786	39,182

For telemetry-driven UAV digital twins, class-conditioned primitive assemblies may provide a bounded, uncertainty-aware, intended pilot-facing obstacle volume when a curated asset is unavailable or too costly to render.

This is a PENDING-EXP claim. It must be evaluated empirically through generation latency, polygon and draw-call budget, dense-scene VR frame rate, operator recognition of class and occupied volume, navigation usefulness under degraded video, unknown-class fallback behavior, and bandwidth comparison against encoded video, telemetry-only boxes, and full-mesh transfer.

15 Preregistered Evaluation Protocol

For a Q1-level validation, SPPA must be evaluated as an operational representation, not as a visual asset generator. The central comparison is against representation choices that a UAV digital twin would realistically use: no rendering, 3D bounding boxes, generic capsules or ellipsoids, billboard labels, curated low-poly assets when available, procedural actors, occupancy style volumes, and SPPA.

A minimal publishable benchmark should include both synthetic scale variants and real or simulated UAV-view detections. Synthetic variants test whether the same semantic graph adapts to different proportions, such as a short truck versus a long truck. UAV-view detections test the harder case: imperfect bounding boxes, partial silhouettes, top-down perspectives, and uncertain yaw. The evaluation must report failure cases, including occlusion, truncated detections, ambiguous front/back orientation, unknown classes, and misleading confidence displays.

16 Threats to Validity

Primitive proxies can misrepresent fine geometry, pose, articulation, and object-specific hazards. A proxy cow or tractor is not a true reconstruction of the observed object. The visual style may also create false confidence: a clean proxy can look more certain than the underlying detector, georeference, or scale estimate. This must be tested with uncertainty visualization rather than assumed.

Top-down UAV viewpoints, occlusion, motion blur, image compression, and partial detections can remove the evidence needed for part placement or signed yaw. The yaw model therefore exposes yaw_ambiguous, but the thresholds and display behavior are PENDING-SPEC. Template bias is another risk: the seven current classes are too small to support broad claims about arbitrary semantic labels.

Language-guided semantic compilation introduces hallucination and certification risk. A language model may propose an implausible part graph for rare or ambiguous labels. SPPA must therefore review language-derived mappings against a finite archetype library, dimension priors,

polygon budgets, and conservative fallback rules before runtime use. This review protocol is PENDING-SPEC.

Finally, runtime performance is hardware- and engine-dependent. Triangle count alone does not predict headset frame rate, draw-call pressure, material cost, or tracking comfort. All dense-scene claims remain PENDING-BENCH.

17 Minimum Work Remaining To Publish

18 Limitations

The current prototype is class-template based and does not infer volumetric scale from a real detection pipeline. It does not consume live YOLO tracks, instance masks, calibrated camera geometry, georeferenced footprints, or uncertainty estimates. It has not been benchmarked inside Unreal with headset frame-rate constraints. The current timings measure a debug OBJ-export path, not a final instanced primitive runtime.

The paper also does not include a pilot or user study. Claims about operator readability, situation awareness, workload, or navigation usefulness remain hypotheses until evaluated. The specified fallback behavior is not implemented in the current prototype and is not yet shown to reduce operational risk.

The next step is to connect the proxy generator to YOLO detections, instance masks or bounding boxes, UAV pose, camera calibration, georeferenced placement, and Unreal actor spawning. The contribution becomes publishable only if it is evaluated under the intended operational constraints: remote piloting, degraded video links, semantic telemetry, and VR real-time rendering.

19 Conclusion

SPPA, as presented here, is a specification for a semantic proxy rendering layer for telemetry-driven UAV digital twins. Static asset libraries cannot cover every possible detected object, and neural image-to-3D systems optimize asset fidelity rather than bounded low-polygon display. Primitive proxy assembly is a plausible operating point for this gap, but the current evidence is limited to a seven-class fixed-template OBJ debug prototype.

The manuscript therefore should not be read as an evaluated UAV/VR system. It defines a runtime contract, descriptor shape, bounded compiler concept, mathematical fitting target, and preregistered evaluation plan. Publication as a full experimental contribution requires the pending implementation, benchmark, fallback, bandwidth, and operator-study artifacts listed above.

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Table 8: Preregistered experimental matrix for testing SPPA.

Experiment	Baselines	Metrics	Required dataset/artifact	Status
Runtime generation	3D box, capsule/ellipsoid, billboard label, curated low-poly asset load, procedural proxy, SPPA	P50/P95 latency, descriptor bytes, triangles, memory	scripted object-spawn benchmark with cold/warm starts	PENDING-BENCH
Dense VR scene	boxes, capsules, billboards, low-poly assets, instanced primitive proxies, detailed meshes	FPS, CPU/GPU frame time, draw calls at 10/50/100/500 entities	Unreal scene and target head-set/hardware profile	PENDING-BENCH
Proportional adaptation	fixed template, bbox-only scale, curated low-poly asset, SPPA	length/width/height error, BEV IoU, volume over/under-estimation, yaw error	synthetic variants plus real or simulated UAV-view detections	PENDING-EXP
Recognition	boxes, capsule/ellipsoid, billboard label, curated low-poly asset, SPPA	class-family accuracy, response time, confusion matrix	randomized operator recognition task	PENDING-USER
Operational navigation	no rendering, boxes, occupancy volume, SPPA	near misses, clearance, task time, NASA-TLX, SAGAT-style situation-awareness probes	degraded-video VR navigation scenario	PENDING-USER
Fallback and uncertainty	no rendering, generic box, uncertainty shell, SPPA fallback	containment rate, false-confidence events, operator interpretation, unknown-class failure rate	unknown or ambiguous class set with occlusions	PENDING-SAFETY
Ablation	SPPA variants without silhouette, footprint, track smoothing, semantic part graph, or uncertainty display	BEV IoU, yaw error, jitter, recognition accuracy	shared benchmark scenes and tracks	PENDING-EXP
Bandwidth	compressed video, telemetry boxes, SPPA descriptors, full meshes	bytes/s, packet rate, retransmission cost, stale-object rate	network replay or controlled link emulator	PENDING-EXP

Table 9: Minimum missing evidence required to avoid the current rejection conditions.

Rejection condition	Missing evidence	Required artifact
Prototype is not the claimed system	No live detections, tracks, calibration, georeferencing, or Unreal spawning	End-to-end Pipeline B replay that creates SPPA descriptors and Unreal actors from recorded or simulated detections (PENDING-IMPL)
No baseline comparison	Boxes, capsules, billboards, low-poly assets, occupancy volumes, and procedural actors are not evaluated	Benchmark scripts and tables comparing latency, geometry, frame rate, and recognition (PENDING-EXP)
No scale-adaptation evidence	Current templates are fixed scale	Dataset with ground-truth or simulated footprints plus BEV IoU, volume error, and yaw error (PENDING-EXP)
No VR runtime evidence	OBJ timing is not headset frame timing	Unreal/VR dense-scene benchmark with CPU/GPU frame time, FPS, draw calls, memory, and entity count scaling (PENDING-BENCH)
No operator evidence	Readability and situation-awareness claims are untested	Controlled user study with recognition, response time, navigation outcome, NASA-TLX, and SAGAT-style probes (PENDING-USER)
No fallback evidence	Conservative display fallback may create false confidence	Unknown-class and ambiguous-detection study with containment, operator interpretation, and failure analysis (PENDING-SAFETY)
No bandwidth evidence	Descriptor cost is not compared with video or meshes	Network replay/link-emulation report with bytes/s, packet rate, staleness, and failure modes (PENDING-EXP)